

Rows Columns and Data Types

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Debugging Strategies

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Trace Tables

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Classify records for Rows columns and schemas. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Rows columns and schemas. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Design a schema independently

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-input case.

6. Solve the given case

Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

Assign intended types and a repair.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A table row is book_id B7, pages 84, available true. Name the type of each field and explain why pages stored as eighty-four complicates calculation.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 7 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Store one book per row and keep reader preferences in a separate record.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-input case.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 State a schema

List the required fields, types and duration boundaries for the worked table.

My reasoning and evidence

13. W2 Classify records

Classify minutes values 0, 60, -1 and 61 under this schema.

My reasoning and evidence

14. W3 Repair a unit problem

A source says 180 seconds but the row says minutes: 180. Give a documented correction.

My reasoning and evidence

15. W4 Design a schema independently

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

My reasoning and evidence

16. W5 Separate validity and truth

A row says Reading, 12 and passes. Does that establish the session really lasted 12 minutes?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Rows Columns and Data Types

A fictional approved book catalogue has three fields per row.

Schema: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False};

C={id:"",pages:80,illustrated:True}.

Rows Columns and Data Types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Column contract

Field names identify values; order is not the only guide.



2. Validate A

B7 is text, 90 is allowed integer, True is Boolean: valid.



3. Validate B

Quoted "90" is text, so pages fails the integer contract.



4. Validate C

Empty id fails even though its other two values have allowed types.

CHANGE ONE THING

Change B's pages from "90" to integer 90 using an approved repair.

Explain the changed result using the matching lesson.

Which two rows originally fail, and do they fail for the same reason?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Publish the recommender's row schema and validation results before ranking books, preserving repair provenance.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Publish the recommender's row schema and validation results before ranking books, preserving repair provenance.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional approved book catalogue has three fields per row.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Empty id fails even though its other two values have allowed types.

1. Column contract
2. Validate C
3. Validate B
4. Validate A

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Validate C Empty id fails even though its other two values have allowed types. A schema makes row meanings and expected types explicit. The string "90" looks like a number when printed but does not meet this strict integer schema. A repair can be authorised later, but must not be assumed silently. Missing or empty identifiers also prevent tracing an item back to its source.

Practice 2: explain the deciding evidence

Changed-case practice: Change B's pages from "90" to integer 90 using an approved repair. Which conclusion is supported by the supplied facts?

1. B now satisfies the declared schema.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: B now satisfies the declared schema. Its id is nonempty and illustrated is Boolean; the page-type defect is removed.

Repair a missing building block

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?. Return to this week and explain what you changed.

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?. Return to this week and explain what you changed.

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If `print(total)` moves before `total=total+2`, what gets printed for the original input?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
